



SURJEET SINGH

AI DEPLOYMENT ENGINEER
& MLOPS SPECIALIST | TEAM LEAD

CONTACT & LINKS

PHONE / WHATSAPP

+91 9198970239

EMAIL

surjeetlko5@gmail.com

LOCATION

Lucknow, UP, India

LINKEDIN

surjeet-singh-a13a4ab2

GITHUB

github.com/surjeetlko

PORTFOLIO

surjeetsingh.vercel.app

TECHNICAL ARSENAL

AI/ops & Infrastructure

Docker MicroK8s Nginx Systemd
Grafana Linux

AI / ML & Vision

YOLOv8/11 EasyOCR Gemini API
OpenCV TensorFlow PyTorch

Backend & Telemetry

Python Flask FastAPI Node-RED
WebSockets MQTT MySQL

EDUCATION

BCA (First Class Distinction)

Sam Higginbottom Univ. of A.T.S (2013–2016)

Percentage: 74.4%

RECOGNITION & CERTS

Google Developer Premium Member

Recognized Member & Author

Advanced Kubernetes: Core

LinkedIn Learning

Machine Learning with Python (V2)

IBM / Credly

NIELIT 'O' Level & CCC

Govt. of India

EXECUTIVE SUMMARY

Mid-level **AI Deployment Engineer, MLOps Specialist & Team Lead** with expertise in taking complex AI/ML models and IoT telemetry systems from development to robust, zero-downtime production environments. Specialized in containerization (Docker & MicroK8s), Edge AI (YOLOv8/11 & Raspberry Pi), Node-RED automation pipelines, and multi-threaded microservices for smart healthcare infrastructure. Recognized thought leader on Google for Developers and Dev.to.

PROFESSIONAL EXPERIENCE

AI Deployment Engineer & Python Developer (Team Lead)

Apr 2024 – Present

Criterion Tech • Lucknow, India

- **28+ Hospital Microservices Migration:** Spearheaded the containerization and zero-downtime migration of 28+ critical hospital microservices (ECG APIs, IoT monitors, Audio DSP engines) into a self-healing Dockerized infrastructure.
- **Generative AI & Vision Pipelines:** Architected end-to-end Computer Vision pipelines (YOLOv8/11 & EasyOCR) integrated with multimodal Google Gemini Pro API for automated clinical medical report generation.
- **High-Concurrency Backend APIs:** Developed robust REST APIs (Flask/FastAPI) and multi-threaded WebSocket servers handling concurrent high-volume IoT telemetry without packet loss.
- **Edge AI Deployment:** Managed Edge AI deployments on Raspberry Pi for resource-constrained medical environments using INT8 quantized models and hardware GPIO alerts.
- **Automated Data Pipelines:** Engineered continuous data acquisition workflows via Node-RED for real-time sensor processing, hospital HIS API integration, and duplicate data suppression.

Senior Technical Trainer & System Administrator

Apr 2016 – Apr 2024

Softech / Aptech Computer Education • Lucknow, India

- **Technical Mentorship & Curriculum:** Delivered advanced training in Python, Database Management, and System Architecture to 500+ students and 15+ instructors with a 90%+ distinction certification rate.
- **Infrastructure & Lab Uptime:** Managed Linux-based laboratory infrastructure, maintaining 99% uptime for practical sessions and real-world student capstone automation projects.

KEY ENGINEERING PROJECTS & SYSTEMS

1. Smart ICU Building Management System (BMS) & Telemetry

Architected a multi-threaded TCP/Flask & WebSocket server platform with ESP32 microcontrollers to monitor critical environmental and power telemetry in ICUs with zero downtime.

2. Edge-AI Medicine Strip & Pill Analyzer

Developed a Computer Vision pipeline using YOLOv8 for pill detection and EasyOCR for text extraction, coupled with Gemini Pro API for generating human-readable medical usage reports.

3. 12-Lead ECG Telemetry & DICOM Ingestion Engine

Built automated data ingestion pipelines processing GE MAC 2000 & Philips TC35 ECG XML/JSON signals with database fallback queues and Orthanc DICOM server archiving.

4. Autonomous Urine & IV-Fluid Medication Monitoring

Engineered camera vision level estimators and Node-RED pipelines calculating hourly patient fluid outputs and medication dosages, synchronizing securely with central hospital APIs.